

SAP Is Equivalent to Lunarity for Coordinatewise-Injective 4×4 Maps

Automated mathematics run `fa_banach_001`
Candidate full result for expert review

22 July 2026

Status. Candidate full solution, likely valid. The converse is a computer-assisted finite proof with exact integer and rational arithmetic. All 14,828 non-lunar symmetry classes have exact norm certificates. Human review and an author-level novelty check are still required.

1 Source problem

Han, Qiu, and Wang ask in Problem 7 of [1], on PDF page 21, whether a map $\Phi : \{1, 2, 3, 4\} \times \{1, 2, 3, 4\} \rightarrow \mathcal{L}$ whose associated Hankel system has SAP must be lunar. In the standing setup of that section, the associated system is defined for coordinatewise-injective maps. The decisive sentence is:

Assume that the Hankel system associated to Φ has SAP. Does it follow that Φ is a lunar map?

Write $[4] = \{0, 1, 2, 3\}$. Coordinatewise injectivity says that every color appears at most once in each row and each column. Put

$$\Gamma_\ell(a, x) = \mathbf{1}_{\{\Phi(a, x) = \ell\}}.$$

For $a, b \in [4]$, define

$$\text{Sol}(a, b) = \{(x, y) \in [4]^2 : \Phi(a, x) = \Phi(b, y)\}.$$

The source paper's equivalent formulation says that Φ is lunar exactly when any two sets $\text{Sol}(a, b)$ and $\text{Sol}(c, d)$ are equal or disjoint.

At matrix level r , for coefficients $C_\ell \in M_r$, set

$$A_C = \sum_{\ell} \Gamma_\ell \otimes C_\ell,$$
$$B_C = \sum_{\ell} (\Gamma_\ell \otimes \Gamma_\ell) \otimes C_\ell.$$

Tensor flips change neither norm nor the SAP question, so this factor order is equivalent to the usual coefficient-first convention. SAP requires $\|A_C\| = \|B_C\|$ for every r and every coefficient family.

Definition 4 (Lunar maps). Consider three sets $\mathcal{A}, \mathcal{X}, \mathcal{L}$. A two-variable map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$ is called a *lunar map* if

- (i) Φ is *coordinatewise injective*, that is, the map $\Phi(a, \cdot)$ is injective on $\mathcal{X}$ for each $a \in \mathcal{A}$ and so is the map $\Phi(\cdot, x)$ on $\mathcal{A}$ for each $x \in \mathcal{X}$.
- (ii) Φ satisfies the *generalized lunar condition*: for any $a, b, c, d \in \mathcal{A}$ and $x, y, z, w \in \mathcal{X}$, the following implication holds:

$$(5.1) \quad (\Phi(a, x) = \Phi(b, y), \Phi(c, x) = \Phi(d, y), \Phi(a, z) = \Phi(b, w)) \implies (\Phi(c, z) = \Phi(d, w)).$$

Given a coordinatewise injective map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$, by the Hankel system associated to the map Φ , we mean the family $\{\Gamma_\ell^\Phi\}_{\ell \in \mathcal{L}} \subset B(\ell^2(\mathcal{X}), \ell^2(\mathcal{A}))$ consisting of operators Γ_ℓ^Φ :

$$(5.2) \quad \Gamma_\ell^\Phi(a, x) = \mathbf{1}(\Phi(a, x) = \ell), \quad (a, x) \in \mathcal{A} \times \mathcal{X}.$$

Note that Φ is coordinatewise injective if and only if the operators $\Gamma_\ell^\Phi \in B(\ell^2(\mathcal{X}), \ell^2(\mathcal{A}))$ are of unit norm $\|\Gamma_\ell^\Phi\| = 1$ for all ℓ in the image of Φ .

Theorem B. For any lunar map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$, the Hankel system $\{\Gamma_\ell^\Phi\}_{\ell \in \mathcal{L}}$ has SAP.

One may wonder whether the lunar condition on a two-variable map Φ is also necessary for the Hankel system $\{\Gamma_\ell^\Phi\}_\ell$ to have SAP. However, even for maps on finite sets of very small size, we are not able to answer this question.

Problem 7. Consider any the map $\Phi : \{1, 2, 3, 4\} \times \{1, 2, 3, 4\} \rightarrow \mathcal{L}$. Assume that the Hankel system of associated to Φ has SAP. Does it follow that Φ is a lunar map?

5.1. Hereditary SAP. In general, the SAP is not preserved by compression (see Example 3.3).

Figure 1: Definition, preceding theorem, and Problem 7 in the source paper, PDF page 21.

2 Main result

Theorem 1. Let $\Phi : [4] \times [4] \rightarrow \mathcal{L}$ be coordinatewise injective. The associated Hankel system has SAP if and only if Φ is lunar.

The implication “lunar implies SAP” is Theorem B of [1]. The new content is the converse.

3 Proof intuition

Regard the 4×4 table as a proper edge-coloring of $K_{4,4}$. A lunar failure is witnessed by two unequal solution sets that overlap. The two row pairs, and independently the two column pairs, have only four possible directed incidence patterns. Keeping all sixteen combined patterns gives a complete normal form without assuming that eight indices are distinct.

An exact restricted-growth recursion then lists every proper coloring and quotients the non-lunar ones by row permutations, column permutations, color relabeling, and transposition. This leaves 14,828 classes. For every class, integer scalar or 2×2 matrix coefficients make an absorbed support block strictly larger than the original operator. Exact Sturm counts put the two squared norms on opposite sides of a rational number.

4 Complete directed normal forms

Lemma 1. Every non-lunar coordinatewise-injective 4×4 table is equivalent under row and column permutations to a table satisfying one of sixteen fixed directed failure patterns.

Proof. Choose unequal overlapping solution sets $\text{Sol}(a, b)$ and $\text{Sol}(c, d)$. After interchanging them if

necessary, choose

$$(x, y) \in \text{Sol}(a, b) \cap \text{Sol}(c, d), \quad (z, w) \in \text{Sol}(a, b) \setminus \text{Sol}(c, d).$$

Thus

$$\begin{aligned} \Phi(a, x) &= \Phi(b, y), & \Phi(c, x) &= \Phi(d, y), \\ \Phi(a, z) &= \Phi(b, w), & \Phi(c, z) &\neq \Phi(d, w). \end{aligned} \tag{1}$$

The ordered row edges (a, b) and (c, d) are distinct non-loops. If they had the same initial vertex, the first two equalities and injectivity in column y would force the same terminal vertex; the solution sets would then be the same. The same argument with column x rules out a common terminal vertex. Consequently, after relabeling rows, the two directed edges have one of the following forms:

type	(a, b, c, d)
R	$(0, 1, 1, 0)$
C_+	$(0, 1, 2, 0)$
C_-	$(0, 1, 1, 2)$
D	$(0, 1, 2, 3)$.

They are respectively reverse, crossed in either orientation, or disjoint.

The points (x, y) and (z, w) are distinct non-loop directed column edges. Because both lie in $\text{Sol}(a, b)$, equality of their initial vertices forces equality of their terminal vertices by row injectivity, and conversely. Hence the same four directed types apply independently to (x, y, z, w) . Substitution of the resulting row and column quadruples into (1) gives the claimed $4 \times 4 = 16$ fixed patterns. $\square$

Retaining both crossed orientations is essential: simultaneous reversal flips both, but does not remove their relative orientation. The verifier therefore uses the redundant sixteen-pattern list directly.

5 Exact classification

The classifier processes the sixteen cells in row-major order. A color is represented by its first-occurrence label. At each cell the recursion either uses an earlier color absent from the current row and column, or introduces the next new label. Every partition of the cells into matchings has exactly one such restricted-growth representation. This proves completeness before any symmetry quotient.

For each table, the code constructs all sixteen sets $\text{Sol}(a, b)$ as bit masks and tests lunarity exactly. For tables satisfying a fixed pattern from Lemma 1, it minimizes the restricted-growth word over all 24^2 row and column permutations, both transpose choices, and canonical color relabeling. The exact census is:

colors	canonical proper	canonical non-lunar	non-lunar orbits
4	24	0	0
5	4,896	4,752	16
6	185,532	183,168	233
7	1,569,864	1,563,168	1,564
8	4,552,617	4,495,344	4,238
9	5,787,840	5,504,064	5,087
10	3,717,416	3,071,652	2,862
11	1,310,592	764,760	736
12	265,110	78,072	87
13	31,152	2,736	5
14	2,076	0	0
15	72	0	0
16	1	0	0
total	17,427,192	15,667,716	14,828

For additional auditability, the sixteen fixed-pattern counts are shown below. They overlap; their union contains 336,208 restricted-growth tables.

row type \ column type	R	C_+	C_-	D
R	0	0	0	37,180
C_+	0	35,439	39,043	34,477
C_-	0	39,043	35,439	34,477
D	37,180	34,477	34,477	42,329

Transposition is a valid orbit symmetry even for matrix witnesses. If C_ℓ witnesses a representative, use $C_\ell^\top$ for its transpose. Then both A_C and B_C are fully transposed, so their norms are unchanged.

6 Exact norm certificates

The support graph of B_C has row vertices (a, b) and column vertices (x, y) , with a block C_ℓ precisely when $\Phi(a, x) = \Phi(b, y) = \ell$. Its connected components are orthogonal row-column blocks. The diagonal component is a copy of A_C . If any other component D_C satisfies

$$\|D_C\| > \|A_C\|, \quad (2)$$

then $\|B_C\| > \|A_C\|$ and SAP fails.

For 7,331 orbit representatives, scalar coefficients suffice. The remaining 7,497 use coefficients $C_\ell \in M_2(\mathbb{Z})$. Candidate coefficients were found numerically, but a record was stored only after the following exact gate passed. For integer matrices $A = A_C$ and $D = D_C$, form

$$p_A(t) = \det(tI - A^\top A), \quad p_D(t) = \det(tI - D^\top D).$$

The record contains a rational q that is not a root of either polynomial. Exact Sturm counts prove

$$\#\{\lambda \in \text{roots}(p_A) : \lambda > q\} = 0, \quad \#\{\lambda \in \text{roots}(p_D) : \lambda > q\} \geq 1.$$

Therefore $\|A\|^2 < q < \|D\|^2$, proving (2) without a floating-point acceptance step.

For example, orbit 0 has representative

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

and scalar coefficients $(1, 1, 1, -1, 1)$. The exact separating threshold is $528077641/1000000000$; the source polynomial has no root above it and the absorbed polynomial has one. Matrix record 86 illustrates the level-two case; its complete integer coefficients, component index, threshold, and root counts are retained in the JSON certificate file rather than rounded for display.

7 Proof of Theorem 1

Proof. If Φ is lunar, its Hankel system has SAP by Theorem B of [1].

Conversely, suppose Φ is non-lunar. Lemma 1 puts a row and column permutation of its table into one of the sixteen directed failure normal forms. The exact enumeration assigns that table, after color relabeling and possibly transposition, to one of the 14,828 orbit representatives. The certificate attached to that representative supplies scalar or 2×2 integer coefficients satisfying (2) exactly. Row and column permutations, color relabeling, and the coefficient-transpose rule transfer the strict norm gap back to Φ . Hence the associated system does not have SAP. This proves the contrapositive SAP implies lunar. $\square$

8 Reproducibility and adversarial review

Run from the packet’s code/ directory:

```
conda run --no-capture-output -n sandbox python \
    verify_full_problem7.py
```

The verifier recompiles the classifier, checks all census constants, checks properness and non-lunarity independently in Python, verifies exact and disjoint certificate coverage, rebuilds the integer block matrices, and repeats every Sturm computation. Its final line is:

PASS: every non-lunar coordinatewise-injective 4x4 map fails SAP

Three discarded routes are important to the audit trail. First, certificates for merge-minimal non-lunar coarsenings do not lift automatically, because tensoring a merged color creates cross terms. Second, a lunar failure need not use four distinct rows and four distinct columns; the earlier disjoint/disjoint census of 6,301 classes was incomplete. Third, quotienting the crossed incidence too aggressively missed a relative-orientation case. The final sixteen-pattern classifier was built after these corrections and all final certificates were remapped or regenerated against its output.

9 Limitations, novelty, and review

This is a computer-assisted exact proof, not a conceptual classification of why each non-lunar pattern violates SAP. It relies on the source theorem for the forward implication.

A bounded novelty search on 22 July 2026 checked arXiv and general web results using the exact title, exact Problem 7 wording, “lunar map”, “Hankel SAP converse”, “prob-hsap”, and the

author names. It found the original arXiv:2407.01048v1 and a 2024 talk announcement mentioning unspecified progress, but no later public theorem resolving Problem 7. Novelty confidence is medium pending contact with the authors.

The result should be reviewed by an operator-space expert and independently audited by someone comfortable with finite classification code. The priority checks are Lemma 1, transposition with matrix coefficients, and random replay of certificate records.

References

- [1] Y. Han, Y. Qiu, and Z. Wang, *Self-absorption of Hankel systems on monoids – a seemingly universal property*, arXiv:2407.01048v1, 2024.